

Interval Based Constrained Path Optimization in Time-Dependent Road Networks

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Nuanced Routing Queries



Most Scenic Route

- Congestion and route attractiveness (e.g., scenic views) are dynamic.
- Users rarely leave at an exact second. They have a window.

Problem Definition

Input:

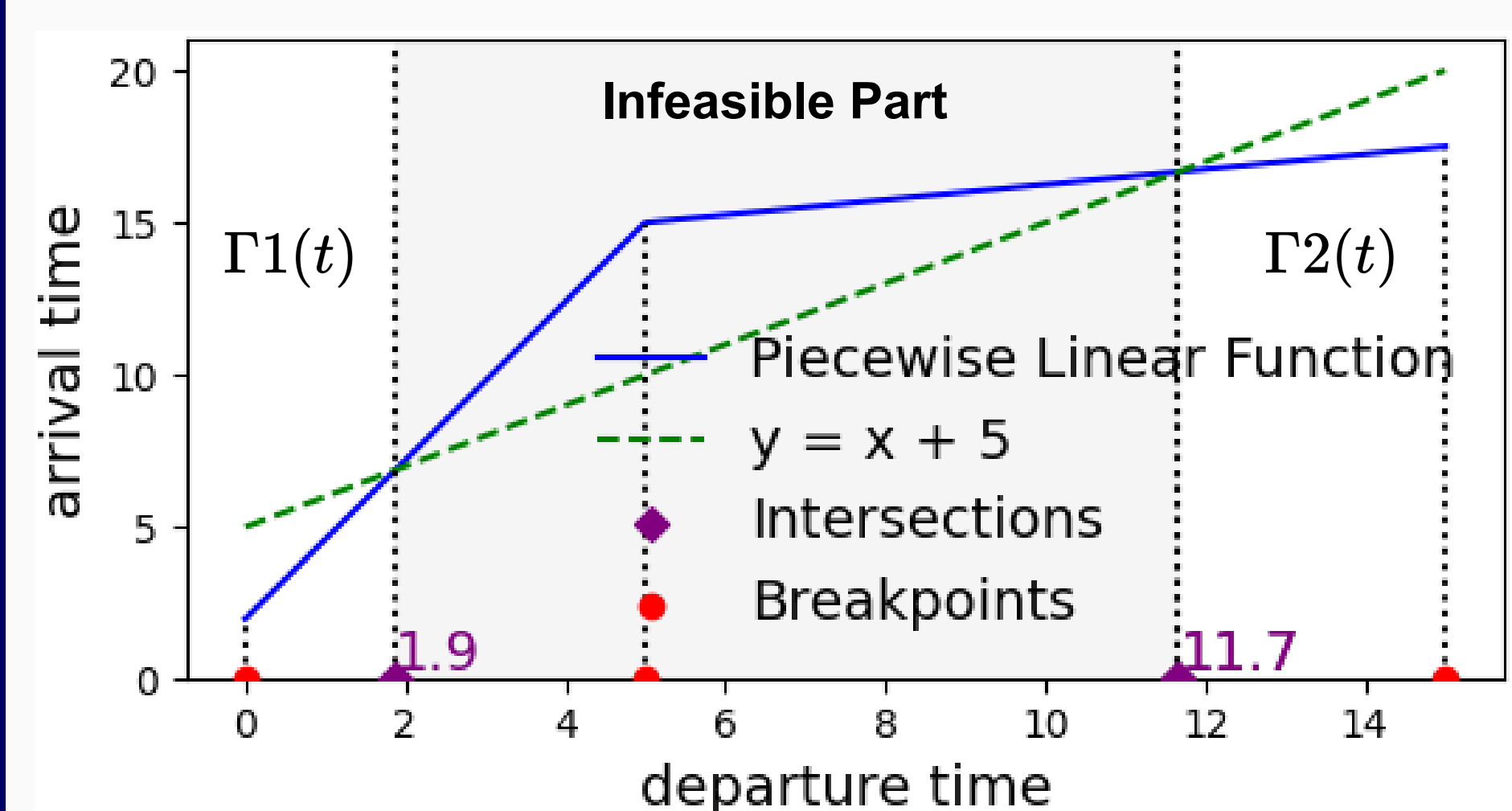
- Input directed graph $G(V, E)$ is a time-dependent graph. Each edge property (score m , travel time c) is time-dependent.
- Source S and Destination D .
- Departure time interval from S : $[t_{dep}^S, t_{dep}^E]$, Budget constrain Θ .

Output: A departure time t_{dep}^* and a path $\mathcal{P}$ from S to D , which maximizes total m and total $c \leq \Theta$ while departing at t_{dep}^* .

iSCOPE

1. Label-based recursive exploration. Each label stores: $\langle v, \Gamma_{curr}, \Phi_{curr}, pred \rangle$ where Γ_{curr} = arrival-time function, Φ_{curr} = accumulated score function.
2. Start from source labels, compose edge functions, and propagate labels.
3. Select best feasible destination label.

Split Function Optimization



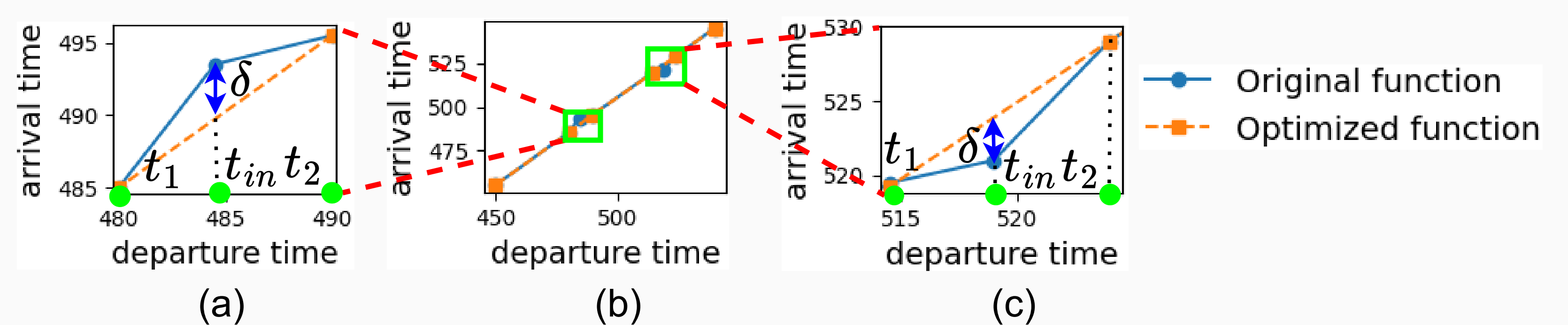
Many function segments violate the time budget.

- Use upper bound $\minTime(v, d)$
- Discard infeasible function intervals early

Contributions

- To the best of our knowledge, we are the first to study the interval TD-CPO problem and develop an algorithm called *iSCOPE*.
- *iSCOPE* constructs *path functions* that capture travel time and score for a range of departure times and optimizes computations by minimizing the function's temporal footprints based on budget constraints.
- *iSCOPE* merges nearby breakpoints, reducing the number of processed breakpoints by over 50% during interval TD-CPO query execution.

Merge Breakpoint Optimizations

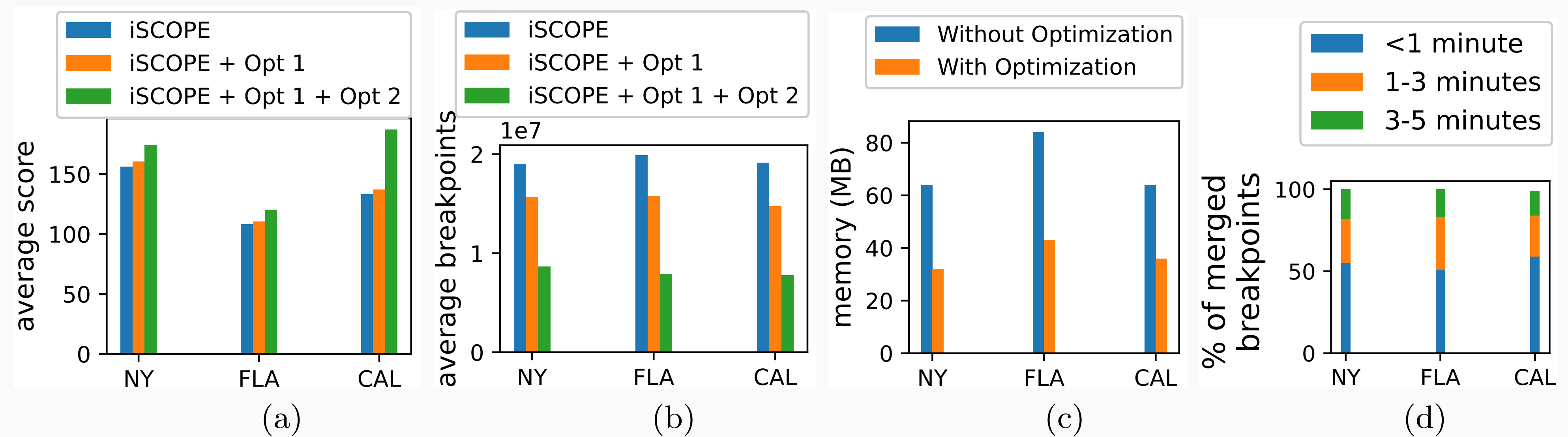


Too many breakpoints increase memory and runtime.

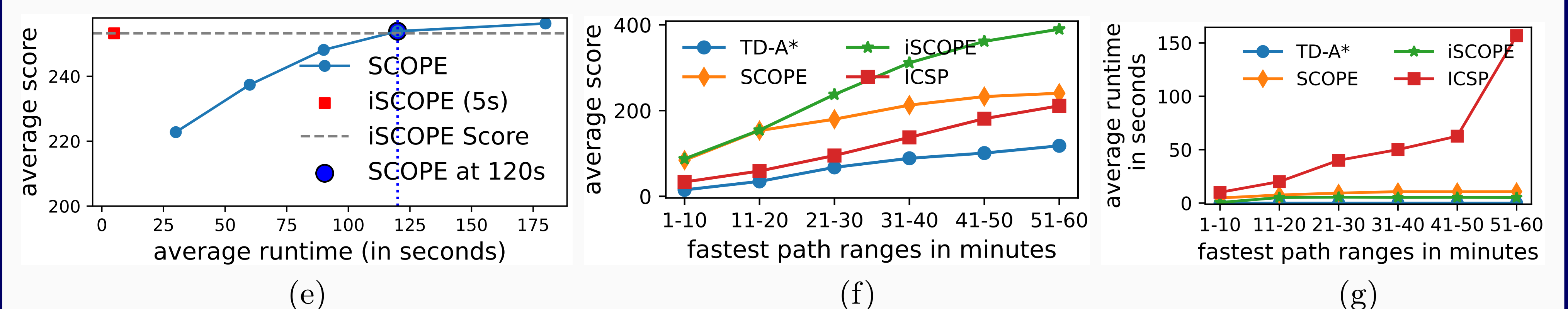
- Cluster nearby breakpoints.
- Merge using probabilistic score assignment.

Experimental Analysis

- **Dataset:** (1) NewYork (NY) (264k nodes), (2) Florida (FLA) (1.07M nodes), (3) California (CAL) (1.89M nodes).
- **Candidate algorithms:** (1) TD-A*[1], (2) SCOPE[2], (2) ICSP[3].
- **Setup:** Intel Xeon Gold 6258R server with 28 cores, 32GB of RAM, a base frequency of 2.7GHz, and a maximum boost frequency of 4GHz.
- **Query set:** Four query bucket with different budget ranges with 100 queries each.



Effect of both the optimizations on *iSCOPE*. (a) average score achieved, (b) average breakpoints processed, (c) average static memory consumed & (d) error caused by merged breakpoints.



(d) Time taken by SCOPE to match the score achieved by *iSCOPE* in 5 seconds, (e) & (f) average score & average runtime comparison in New York network for all the candidates.

References

- [1] Ding, B., et al.: *Finding time-dependent shortest paths over large graphs*. International Conference on Extending Database Technology (2008)
- [2] Dutta, K.K., et al.: *Constrain path optimization on time-dependent road networks*. International Conference on Web Information Systems Engineering (2024)
- [3] Ye Yuan et al.: *Constrained Shortest Path Query in a Large Time-Dependent Graph*. Proc. VLDB Endow. (2019)

Acknowledgements



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